Mahmoud Alnabawi

Data Analyst — Business Intelligence — Quality Analytics

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Professional Summary

Data Analyst with 3+ years of experience working with manufacturing, quality, laboratory, and operational data. Skilled in collecting, cleaning, transforming, analyzing, and visualizing data to identify trends, quality issues, process deviations, and performance improvement opportunities. Hands-on experience with Microsoft Excel, Power Query, Power BI, DAX, SQL, and Python for data preparation, exploratory data analysis, statistical analysis, KPI monitoring, dashboard development, and reporting. Experienced in applying data-driven problem solving, Root Cause Analysis, Pareto Analysis, Lean Six Sigma, and continuous improvement within manufacturing and quality environments. Currently developing advanced Data Analytics and Business Intelligence skills through the Digital Egypt Pioneers Initiative (DEPI) – Data Analytics for Professionals, supported by professional Data Science and Advanced Data Analytics programs on Coursera.

Core Data Analytics Skills

- **Data Analysis:** Data Cleaning, Data Transformation, Data Validation, Exploratory Data Analysis (EDA), Trend Analysis, Descriptive Statistics, KPI Analysis
- **Business Intelligence:** Power BI, Dashboard Development, Data Visualization, KPI Monitoring, Interactive Reports, Data Storytelling
- **Excel:** Advanced Excel, Pivot Tables, Power Query, Lookup Functions, Conditional Analysis, Charts, Reporting, Data Cleaning
- **SQL:** SELECT, WHERE, GROUP BY, ORDER BY, JOINs, Aggregations, Subqueries, Data Extraction
- **Python:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Data Cleaning, EDA and Visualization
- **Power BI / DAX:** Data Modeling, Measures, Calculated Columns, KPI Calculations, Time Intelligence, Interactive Dashboards
- **Statistical & Quality Analytics:** Statistical Analysis, Pareto Analysis, Correlation Analysis, Root Cause Analysis, Process Performance, Six Sigma

Professional Experience

Mac Carpet – Egypt

Quality Assurance & Laboratory Specialist

- Analyze manufacturing, quality, laboratory, raw material, and testing datasets to identify trends, deviations, recurring defects, and process improvement opportunities.
- Collect, clean, validate, organize, and consolidate data from multiple internal and external sources while maintaining data accuracy, consistency, traceability, and integrity.
- Use Microsoft Excel and analytical techniques to prepare quality reports, performance summaries, KPI reports, and management insights.
- Apply exploratory data analysis, trend analysis, Pareto analysis, and statistical thinking to identify major contributors to quality problems and process deviations.
- Monitor quality and testing KPIs and analyze performance trends to support data-driven decision making and corrective actions.

- Investigate quality deviations using structured data analysis, Root Cause Analysis, and Lean Six Sigma problem-solving methodologies.
- Consolidate and analyze laboratory testing results from internal departments and external laboratories to support technical decisions.
- Work with large volumes of laboratory, quality, supplier, and manufacturing records while maintaining data quality and documentation standards.
- Transform operational and quality data into actionable insights that support continuous improvement, defect reduction, and process performance.
- Collaborate with Production, R&D, Quality, Suppliers, and external laboratories to support data-driven investigations and improvement initiatives.

Data Analytics Projects

Manufacturing Quality and Defect Analysis

- Performed exploratory analysis of manufacturing quality data to identify defect patterns, recurring issues, and major contributors to overall defect rates.
- Applied data cleaning, aggregation, trend analysis, and Pareto analysis to prioritize the most significant quality issues.
- Developed analytical reports and visualizations to communicate quality trends and support corrective and preventive actions.
- Used Excel and statistical analysis techniques to transform raw quality records into meaningful performance insights.

Manufacturing KPI Dashboard – Power BI

- Developed interactive Power BI dashboards for monitoring manufacturing and quality performance indicators.
- Used Power Query to clean, transform, and prepare raw data from multiple sources.
- Created DAX measures and calculated fields for KPI calculations, performance analysis, and trend monitoring.
- Designed interactive visualizations to support management reporting and data-driven decision making.

IBM Data Science and Coursera Projects

- Completed hands-on projects involving data cleaning, exploratory data analysis, data visualization, SQL, Python, and real-world datasets.
- Used Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn for data analysis and visualization.
- Applied SQL queries to extract, filter, aggregate, join, and analyze structured datasets.
- Developed data-driven insights and visualizations to communicate analytical findings.

Education

Bachelor of Science in Chemistry
Faculty of Science, Al-Azhar University

Professional Development

Digital Egypt Pioneers Initiative (DEPI) – Data Analytics for Professionals Currently Studying

- Advanced Data Analytics, Business Intelligence, statistical analysis, data visualization, and data-driven decision making.

- Practical application of analytical tools and methodologies to business and operational problems.

IBM Data Science Professional Certificate Coursera

- Python for Data Science and AI
- Databases and SQL for Data Science
- Data Analysis with Python
- Data Visualization with Python
- Machine Learning with Python
- Applied Data Science Projects

Google Advanced Data Analytics Coursera

- Advanced data analysis, statistical analysis, Python programming, data visualization, and data-driven problem solving.

Lean Six Sigma Black Belt Coursera

- DMAIC, process improvement, statistical thinking, Root Cause Analysis, process performance, and data-driven continuous improvement.

Technical Skills

Data Analysis:	Excel, Python, Pandas, NumPy, SQL, Exploratory Data Analysis
Business Intelligence:	Power BI, Power Query, DAX, Data Modeling, Dashboard Development
Data Visualization:	Power BI, Matplotlib, Seaborn, KPI Dashboards, Data Storytelling
Databases:	SQL, Relational Databases, Data Extraction, Data Aggregation
Statistics:	Descriptive Statistics, Correlation Analysis, Trend Analysis, Pareto Analysis, Statistical Analysis
Quality Analytics:	Root Cause Analysis, Six Sigma, Process Performance, Defect Analysis, Continuous Improvement
Tools:	Microsoft Excel, Power BI, Python, Jupyter Notebook, SQL, Power Query

Additional Professional Skills: Analytical Thinking and Problem Solving, Data-Driven Decision Making, Business and Process Understanding, Manufacturing and Quality Analytics, Technical Reporting and Data Storytelling, Cross-functional Collaboration, Attention to Detail, Continuous Learning

Extracurricular Activities

- **Volunteer, Charitable Medical Organizations (Egypt)** — Contribute time and effort supporting community medical initiatives and outreach programs.
- **Arabic-Language Data Analytics Content Creator** — Create and script educational video content in Arabic covering Python and SQL fundamentals, aimed at helping Arabic-speaking learners build data analysis skills.
- **LinkedIn Personal Branding** — Actively publish posts on quality engineering, data analytics, and professional certifications to build a personal brand and share technical knowledge with a professional network.
- **ECRAIL Program, AMIDEAST Egypt (U.S. Embassy-supported)** — Enrolled in a structured program developing Python/Pandas and SQL skills alongside professional peers.